# Smart Earring with Heart Rate Monitoring

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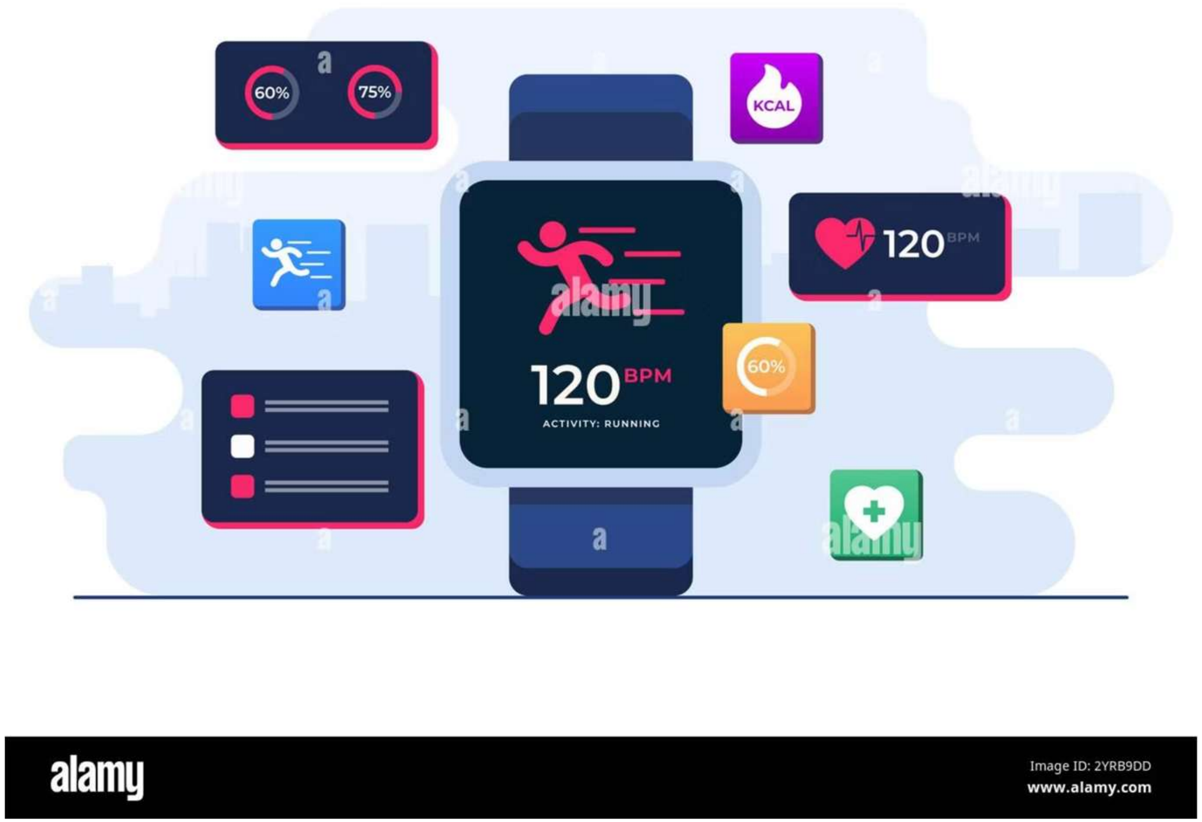
# Introduction

This project presents a smart earring system that monitors heart rate in real-time using pulse oximetry. The device connects via Bluetooth to smartphones, enabling continuous cardiovascular health tracking.

The compact design integrates medical-grade sensors into fashionable jewelry, making health monitoring seamless and accessible.



# Why Smart Earrings?

### Continuous Monitoring

Track heart health throughout daily activities without disrupting routine

### Fashion Meets Function

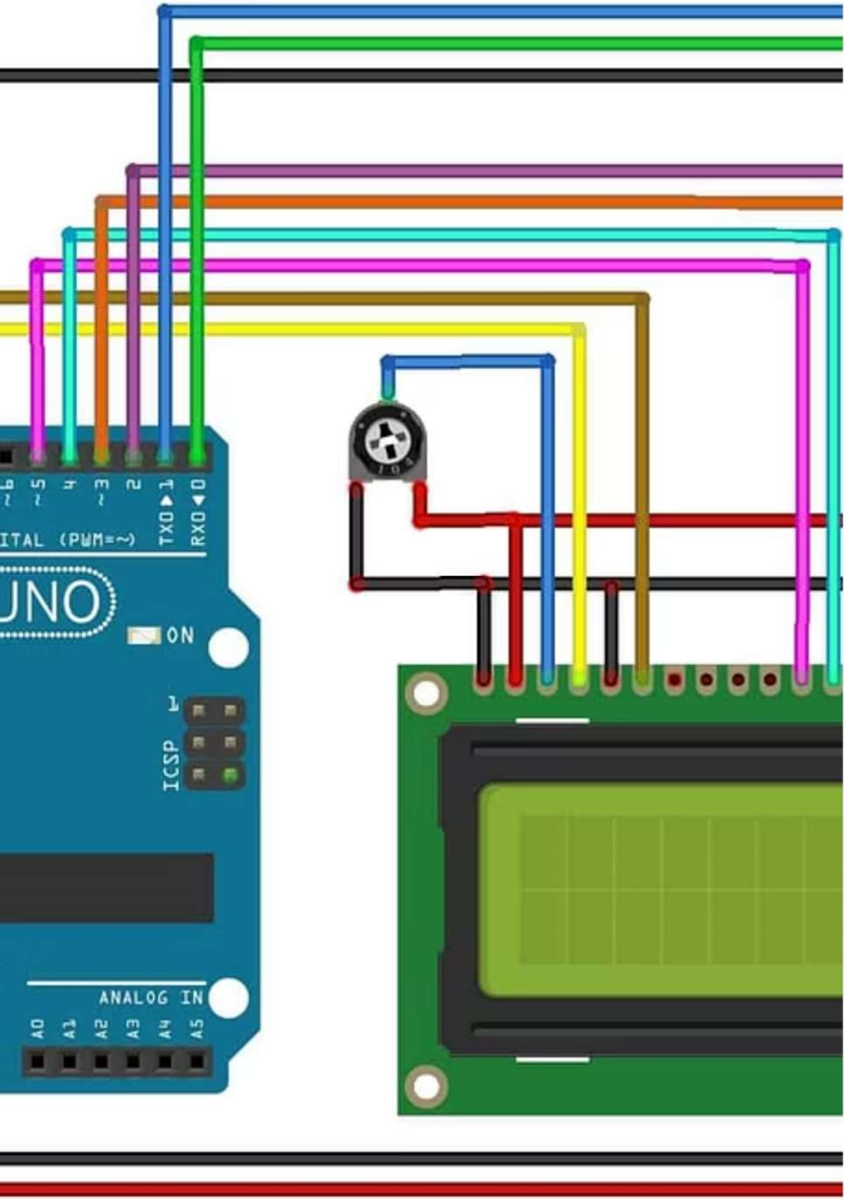
Discreet wearable design that complements personal style

### Smartphone Integration

Real-time data visualization and alerts on mobile devices

### Health Alerts

Immediate notifications for abnormal heart rate patterns




# System Block Diagram

01

## Pulse Sensor

Optical sensor detects blood flow variations

02

## Microcontroller

Processes analog signals and calculates heart rate

03

## Bluetooth Module

Wireless transmission to smartphone application

04

## Mobile App

Data visualization, storage, and notifications

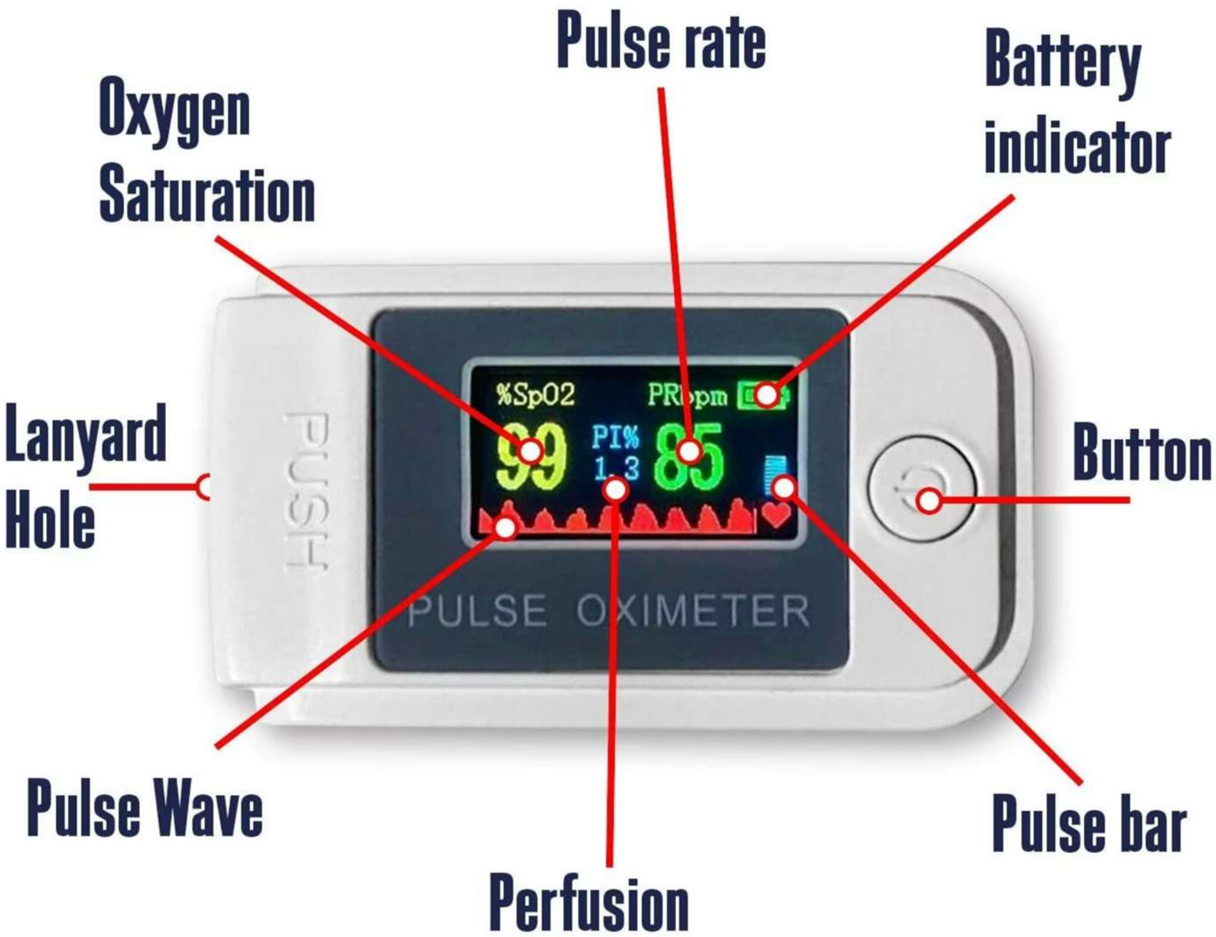
# How It Works

## Signal Acquisition

- Infrared LED emits light through earlobe tissue
- Photodiode detects light absorption changes from blood flow
- Pulse oximetry principle measures oxygen saturation

## Data Processing

- Analog-to-digital conversion of pulse signals
- Signal filtering removes noise and motion artifacts
- Algorithm calculates beats per minute (BPM)



# Hardware Components

1

### MAX30100 Pulse Sensor

Integrated pulse oximeter and heart rate monitor IC with built-in LEDs and photodetector

2

### Arduino Nano

Microcontroller unit processes sensor data and manages Bluetooth communication

3

### HC-05 Bluetooth Module

Serial communication module enables wireless data transfer to smartphones

4

### 3.7V Lithium Battery

Compact power source providing 8-12 hours of continuous operation

5

### PCB & Enclosure

Custom printed circuit board with waterproof casing for ear attachment

# Budget Analysis

**₹1,250**

### Total Cost

Approximate component expenses

**₹650**

### MAX30100 Sensor

Primary sensing component

**₹250**

### Arduino Nano

Processing unit

**₹170**

### HC-05 Module

Bluetooth connectivity

**₹170**

### Battery & PCB

Power and assembly

All components are readily available from electronics suppliers, making this project feasible for undergraduate implementation within typical laboratory budgets in Indian Rupees (INR).

# Advantages & Challenges

## ✓ Advantages

- Non-invasive, continuous health monitoring
- Compact and fashionable design
- Real-time data with smartphone alerts
- Low power consumption for extended use
- Affordable components and easy assembly
- Comfortable for all-day wear

## ⚠ Limitations

- Requires stable ear attachment during movement
- Battery life limited to 8-12 hours
- Ambient light may interfere with readings
- Not a medical-grade diagnostic device
- Water resistance requires additional sealing
- Signal accuracy depends on proper fit


# Applications & Use Cases

### Fitness Tracking

Athletes and gym-goers monitor cardiovascular performance during workouts

### Elderly Care

Remote heart health monitoring for seniors with cardiovascular conditions

### Medical Screening

Preliminary screening for arrhythmias and abnormal heart rhythms

### Stress Management

Track heart rate variability as stress indicator during daily activities

# Conclusion & Future Scope

## Project Summary

This smart earring system successfully demonstrates wearable health technology integration using affordable, accessible components. The device provides real-time heart rate monitoring with smartphone connectivity.

## Future Enhancements

- Integration with AI for predictive health analytics
- Additional sensors for blood pressure and temperature
- Extended battery life with energy harvesting
- Medical certification for clinical applications

**References:** MAX30100 datasheet, Arduino Bluetooth libraries, pulse oximetry research papers